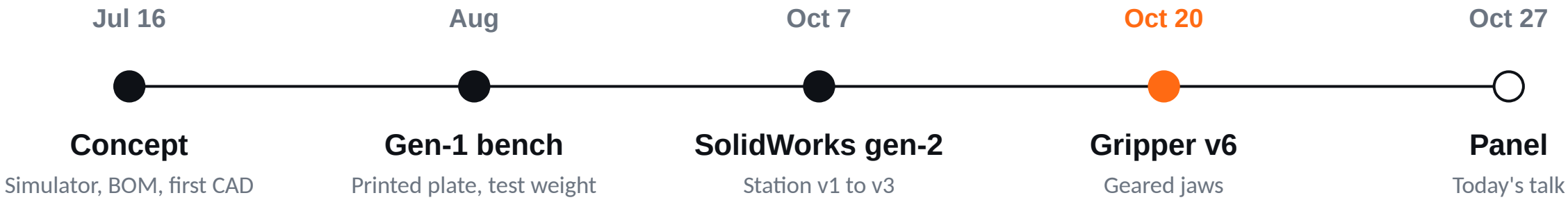
RoomCleaner

A four-cable ceiling robot that picks laundry off the floor

Jasper Buntinx · Technical presentation · Oct 27, 2026



Solo build: four winch stations, a camera claw, and the software.



BOUGHT

A4988 driver, ESP32, DROK buck, MG996R servo.

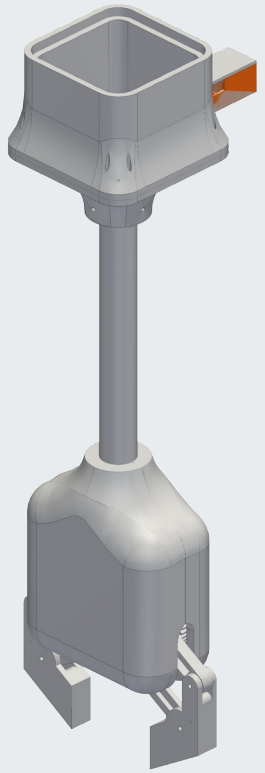
DESIGNED, PRINTED, MACHINED, WIRED BY ME

Everything else.

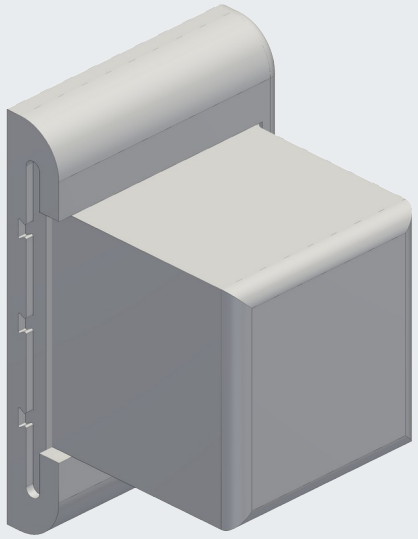
HOW TO READ THE TAGS

MEASURED	On hardware, with an instrument
CALCULATED	Hand calculation, stated inputs
SIMULATED	Run in a model or simulator
PLANNED	Not done yet

Done means jeans from anywhere on the floor into the hamper.



Claw v6 with camera bay

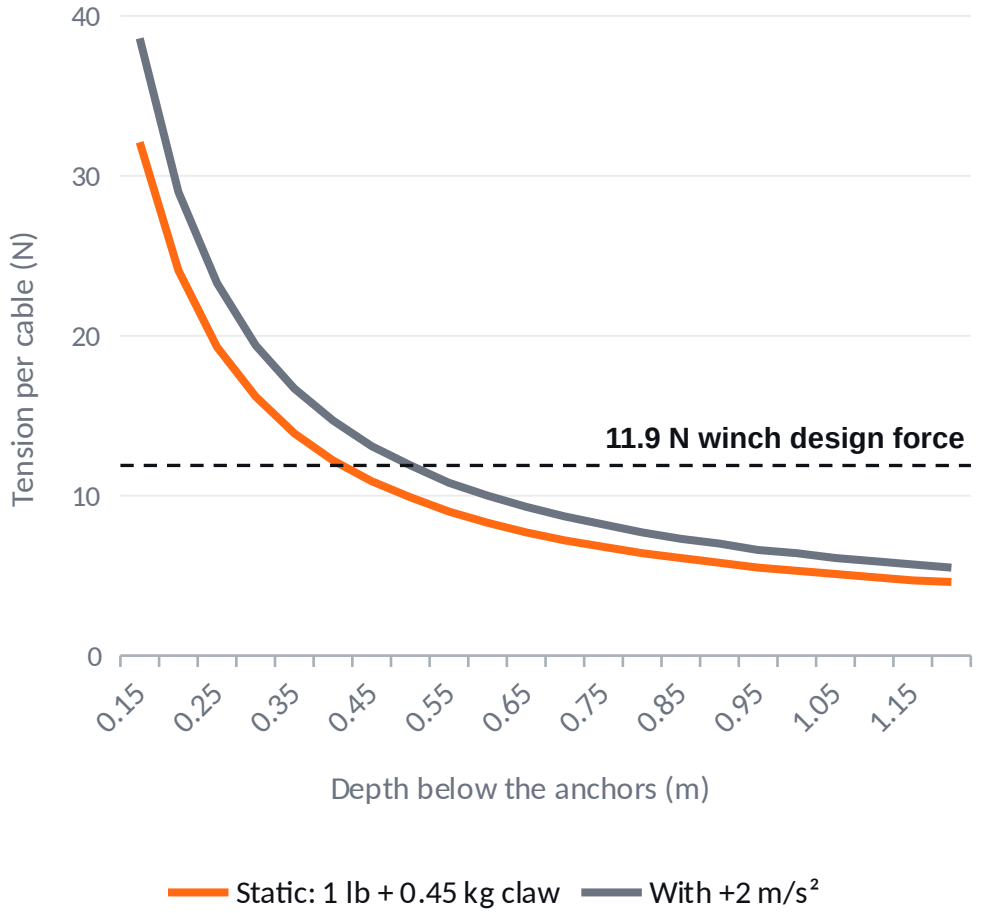


Winch station v3

0.9 kg	Payload of jeans, plus 2 m/s² acceleration
≤ 450 g	Claw mass budget
4 × studs	Winch stations screwed into wall studs
1 switch	The switched power strip is the only kill switch
0 people	Nobody under the robot while it moves

CALCULATED

Cable angle, not payload, sets the limits; walls bind before the ceiling.



1	WANTED	Use the whole room.
2	REALIZED	**Tension: 32 N at 0.15 m depth, 8.3 N at 0.6 m.**
3	DECIDED	Keep a 0.4 m ceiling margin plus a wall margin.
4	RESULT	At task heights, corners bind: 1.1 × total weight.
5	BUT	The claw's own mass counts exactly like payload.

FUNDAMENTALS

$$T = W / (4 \sin \varphi),\ \sin \varphi = d / \sqrt{(d^2 + D_h^2)}$$

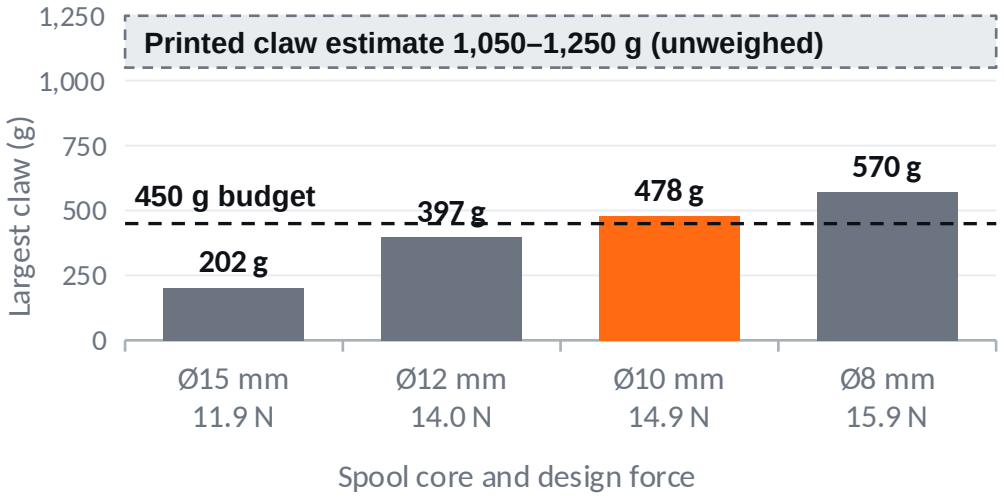
W = 8.86 N (1 lb + 0.45 kg claw), D_h = 2.17 m: T = 32.1 N at d = 0.15 m, 8.3 N at d = 0.60 m.

MEASURED
CALCULATED

The driver, not the motor, capped my pull; geometry buys it back.

0.85 V → 1.06 A

Vref cap on my A4988 boards, and the current it allows



PLANNED — result after the Oct 22 test

Luggage-scale pull at 1.06 A, empty and full spool; replaces the assumed 0.7 running-torque factor.

1	WANTED	Raise drive current from 1.0 to 1.35 A for more pull.
2	REALIZED	**Vref pot maxes at 0.85 V: current caps at 1.06 A.**
3	DECIDED	Run at the cap; recover pull with a smaller spool core.
4	RESULT	Winches hold about 1.1 kg; claw plus jeans is about 2 kg.
5	BUT	The 0.7 running-torque factor is assumed; a pull test decides.

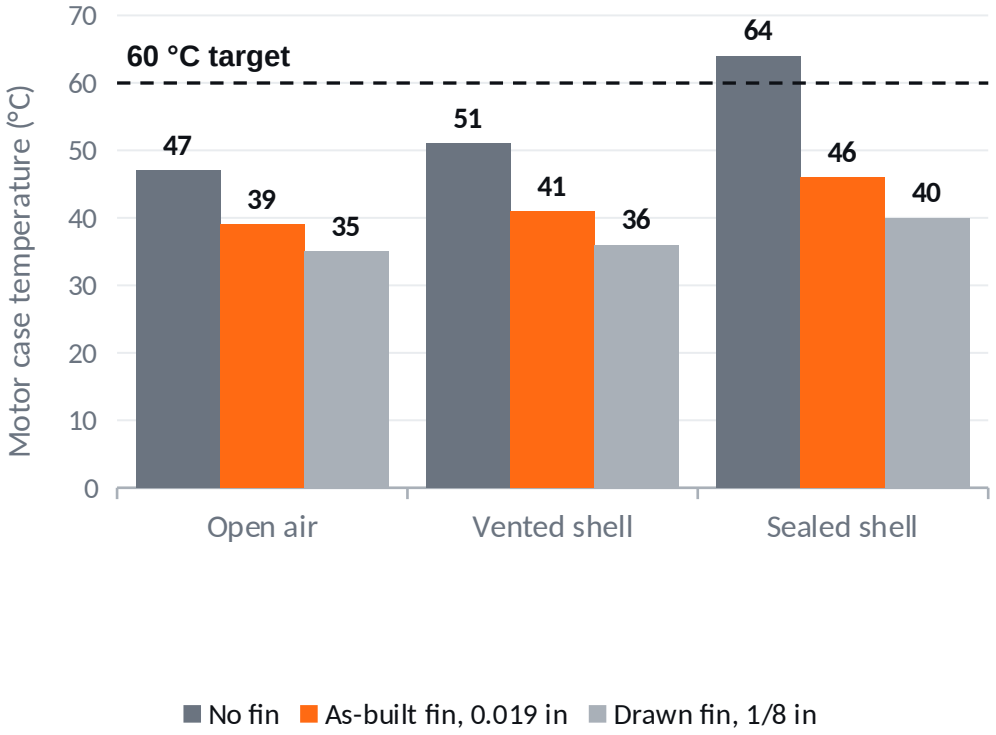
FUNDAMENTALS

$$I = V_{ref} / (8 \cdot R_s) = 0.85\ V / 0.8\ \Omega = 1.06\ A$$

$T = \tau / r$: $\tau_{run} = 0.7 \times 0.255\ N{\cdot}m = 0.179\ N{\cdot}m \rightarrow 23.8\ N$ at $r = 7.5\ mm$, $17.9\ N$ at a full spool ($r = 10\ mm$).

CALCULATED

Sealing the station made the motor mount its hottest, most loaded joint.



PLANNED — result after the Oct 22 test

Hold test at 1.06 A with thermometers on the motor case, fin and PETG tab; pass limit below 60 °C.

1 **WANTED** — A clean, closed printed shell around each winch.

2 **REALIZED** — A holding stepper dissipates I^2R; PETG conducts 1/800 as well as aluminium.

3 **DECIDED** — Vents plus an aluminium fin through the shell.

4 **RESULT** — **Sealed, 1.06 A: 64 °C bare, 46 °C with the fin.**

5 **BUT** — No thermometer reading yet; the hold test is planned.

FUNDAMENTALS

$$P = I^2R; \quad R_{cond} = L / (k\,A); \quad R_{conv} = 1 / (h\,A)$$

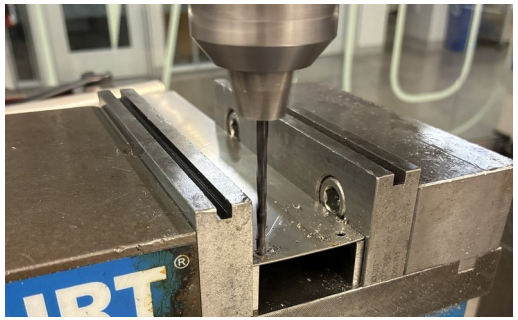
$P = 1.06^2\ A^2 \times 1.8\ \Omega = 2.0$ W (R assumed). Flange path: PETG 14 K/W vs aluminium 0.011 K/W, 1260 × worse.

MEASURED
CALCULATED

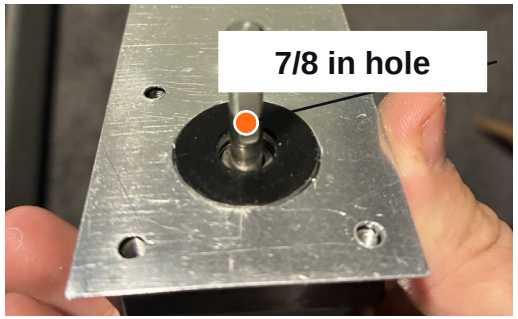
I drew 1/8-inch, built 0.019-inch, then re-checked the physics.



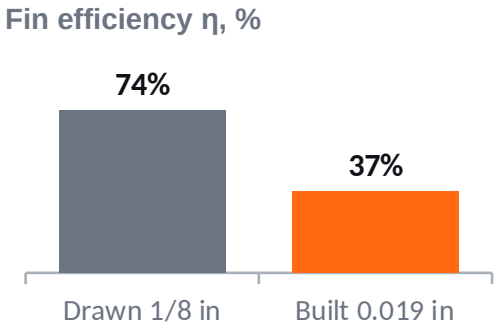
1 Sheared blanks beside my drawing



2 Drilling the holes on the mill



3 Fitted: boss centred, 4 holes line up



PLANNED — result after the Oct 22 test

Hold test: does the thin sheet bow or buzz, and are the screw threads deep enough?

1 **WANTED** — The drawn 1/8 in plate: fin efficiency about 74 %.

2 **REALIZED** — The sheet I could shear and drill was 0.019 in.

3 **DECIDED** — **Re-run the fin equation first: efficiency falls 74 % to 37 %.**

4 **RESULT** — Fin still carries about half the heat: 46 °C sealed.

5 **BUT** — Thin sheet may buzz; screws reach 2.7 mm deeper, so check threads.

FUNDAMENTALS

$$\eta = \tanh(mL) / (mL), \quad m = \sqrt{(2h / (k\,t))}$$

t = 0.483 mm: m = 13.5 /m, mL = 2.69, η = 37 %, R_fin 10.9 → 22 K/W. Drawn 1/8 in: mL = 1.05, η = 74 %.